CORNERSTONE MINOR EVALUATION

SCHOOL OF ENGINEERING
MATHEMATICS AND COMPUTING

AI BASED PLAGIARISM FRAMEWORK

Minor Evaluation

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PROBLEM STATEMENT

- Generative AI has transformed education by helping students write reports, assignments, and research content.
- Along with its benefits, AI has increased the risk of academic plagiarism through copied, paraphrased, and AI-generated content.
- Existing plagiarism checkers mainly detect direct copying but often struggle with semantic paraphrasing and AI-generated text.
- Our project proposes an AI-powered framework to help maintain academic integrity in universities.

Build a plagiarism detection system that analyzes uploaded documents and identifies multiple forms of plagiarism using AI and NLP techniques.

OBJECTIVE

1. Detect Direct Plagiarism by comparing copied text.
2. Detect Paraphrased Plagiarism using semantic similarity.
3. Analyze AI-Generated Text characteristics.
4. Design a framework that can work in real-world academic scenarios.
5. Provide a tool usable by: Students, Faculty, Professors, Researchers, & University Administration.

Expected Outcome:

A unified plagiarism analysis report with similarity score and highlighted suspicious content.

WORKFLOW

Document Upload
PDF / DOCX / TXT

Preprocessing
Text extraction, cleaning, tokenization, normalization.

AI Detection Modules
Direct Matching
Semantic Similarity
AI Text Analysis

Similarity Analysis
Sentence similarity and suspicious passage detection.

Final Report
Similarity percentage and highlighted plagiarized content

Technology Used : Flask, NLP, Python, Cosine Similarity, Sentence Transformers.

IMPLEMENTATION

Module	Status
Document Upload Interface	Pending
Text Extraction & Preprocessing	Pending
Direct Plagiarism Detection	Partialy Done
Paraphrase Detection	Research Phase
AI Text Detection	Research Phase
Report Generation	Pending

Currently working on v0.1

- Accept academic documents.
- Process and clean extracted text.
- Compare text similarity.
- Highlight potentially copied passages.
- Generate an initial plagiarism report.

CONCLUSION

- Developed a foundational AI-based plagiarism detection framework.
- Combined NLP preprocessing with plagiarism analysis modules.
- Focused on supporting academic integrity rather than replacing human evaluation.
- The framework is scalable for future deployment in universities.

REFRENCES

Andreas C. Müller and Sarah Guido Introduction to Machine Learning with Python A Guide for Data Scientists

Nithin Buduma, Nikhil Buduma, and Joe Papa with contributions by Nicholas Locascio Fundamentals of Deep Learning Designing Next-Generation Machine Intelligence Algorithms

Creating a Plagiarism Detector with Python: A Comprehensive Guide.

Plagiarism Detection using Python

Thank You